

**FORECASTING COVID-19 WITH
TEMPORAL HIERARCHIES AND ENSEMBLE METHODS**

A Thesis Presented

by

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ABSTRACT

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Infectious disease forecasting efforts underwent rapid growth during the COVID-19 pandemic, providing guidance for pandemic response and about potential future trends. Yet despite their importance, short-term forecasting models often struggled to produce accurate real-time predictions of a complex and rapidly changing system. This gap in accuracy, which persists three years into the pandemic, warrants the exploration and testing of new methods to glean fresh insights.

In this work, we examine the application of the temporal hierarchical forecasting (THieF) methodology to probabilistic forecasts of COVID-19 incident hospital admissions in the United States. THieF is a recent, innovative forecasting technique that aggregates time-series data into a hierarchy made up of different temporal scales, produces forecasts at each level of the hierarchy, then reconciles those forecasts using optimized weighted forecast combination. While THieF’s unique approach has shown substantial accuracy improvements in a diverse range of applications, such as opera-

tions management and emergency room admission predictions, this technique has not previously been applied to outbreak forecasting.

We generate candidate models formulated using the THieF methodology, which differ by their hierarchy schemes and data transformations, and weighted ensembles of the THieF models. The models are validated using weighted interval score (WIS), and the top-performing subset is compared to several benchmark models. These models include simple SARIMA models, a naive baseline model, and a median ensemble of operational incident hospitalization models from the US COVID-19 Forecast Hub. The THieF models and THieF ensembles demonstrate improvements in WIS and MAE, as well as competitive prediction interval coverage, over the benchmark models for both the validation and testing phases. These accuracy improvements suggest the THieF methodology may serve as a promising new technique for infectious disease forecasting that makes better predictions in real-time.

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CHAPTER 1

INTRODUCTION

- 1.1 Infectious disease forecasting**
- 1.2 COVID-19 forecasting**
 - 1.2.1 US COVID-19 Forecast Hub**
- 1.3 Temporal hierarchical forecasting**
- 1.4 Goals**

CHAPTER 2

METHODS

2.1 Surveillance data

This analysis uses confirmed hospital admissions from HealthData.gov from the COVID-19 Reported Patient Impact and Hospital Capacity by State Time Series. This dataset reports state-aggregated daily incident hospitalizations starting on January 1, 2020 for U.S. states and territories. The data come from three primary sources: HHSTeleTracking, state/territorial health departments which report to HHS Protect, and National Health Care Safety Network (prior to July 15, 2020) [7].

On December 1, 2020, the U.S. COVID-19 Forecast Hub designated this dataset as the official ground truth data for COVID-19 incident hospitalizations. (No other sources were designated official ground truth data prior to this date.) [2] This meant that the HealthData.gov data was used to evaluate the performance of real-time incident hospitalization forecasts submitted to the Forecast Hub. Hence, teams were encouraged to use this hospitalization data set to train their models [3]. In the analysis, we also use this dataset as both input training data for our models and truth data against which to perform accuracy evaluations.

Hospitalization reporting was sporadic for the first four months of the pandemic, with consistent, reliable values for major states and territories achieved only in late July 2020. Specifically, reports of daily confirmed hospitalizations for all 50 states, Washington D.C., Puerto Rico, and the U.S. as a whole began on July 27, 2020. Although COVID-19 hospitalization reporting stabilized after that for cases and deaths, hospitalization values are often more reliable due to requirements for Medicare reim-

bursement. COVID-19 cases have been vastly underreported for the entire duration of the pandemic [5], and this problem has worsened with dwindling testing. Further, truth data collection and reporting for COVID-19 incident cases and deaths has been discontinued by Johns Hopkins University (the official ground truth data source) as of March 10, 2023 [4].

Surveillance data, like the HealthData.gov confirmed hospitalizations, sometimes required revisions due to reporting anomalies. Hospitalizations might be entered on incorrect dates or added after COVID-19 was determined to be the cause of hospitalization. Updates to values usually occurred within the first week or so of posting. As a result, real-time forecast models might train on data that include incident hospitalization numbers for location-date combinations that do not reflect the finalized values.

We query these confirmed hospitalizations from the ‘covidData’ package through the ‘covidHubUtils’ package, both developed by Reich Lab at the University of Massachusetts Amherst to deal with COVID-19 surveillance data. Versions of this data are stored such that users may access the truth data as of a specified date. For example, it is possible to retrieve of version of the confirmed hospitalization data set as of July 27, 2020. To ensure fair comparisons between the models developed for this analysis and some real-time models, our models are trained on the confirmed hospitalizations that would have been available as of the ”forecast date” of the retrospective forecast.

2.2 Forecasts

2.2.1 Forecast quantiles

The Forecast Hub specifies the usage of 23 quantiles

$$\mathbb{Q} = \{.010, 0.025, .050, .100, \dots, .900, .950, .990\}$$

to represent a full probabilistic distribution, with the median 0.500 quantile usually taken as the model’s point forecast. We follow this construction, though with a slight modification: prediction intervals are used to define a full distribution of forecasts. The bounds of these intervals can be used to obtain the desired quantiles, as shown in the table below. For example, the values for a 0.500 and a 0.950 quantile (see the Quantile row) are obtained by taking the lower and upper bounds of the 95% prediction interval (see the Interval row, where "L" and "U" represent the lower and upper bounds, respectively).

Quantile	.010	.025	.050	0.100	0.150	...	0.500	...	0.900	0.950	.975	.990
Interval	98L	95L	90L	80L	70L	...	Median	...	80U	90U	95U	98U

Table 2.1. Table of the 23 quantiles required by the Forecast Hub and their associated prediction interval bound.

2.2.2 Forecast locations

We make forecasts for 53 of 55 locations from the HealthData.gov confirmed hospitalizations dataset during the period of analysis. These locations include the 50 states, four jurisdictions/territories (Washington D.C., Puerto Rico, American Samoa, and Guam), and a U.S. national level. American Samoa and Guam are excluded due to their very low or zero COVID-19 incident hospitalizations over the entire pandemic. Forecasts for low-count locations that predicted single-digit or zero hospitalizations were often very accurate; hence, these locations do not offer meaningful contributions to our understanding of differences in forecast skill between modeling approaches [3].

2.2.3 Forecast horizons

2.3 Date range of analysis

2.4 Benchmark comparison models

Two models from the Forecast Hub—the COVIDhub-baseline and COVIDhub-4_week_ensemble—are selected to serve as comparisons for the models in this analysis during the testing phase evaluation. Many of the metrics used in this evaluation are not especially meaningful on their own, requiring a relative comparison among models forecasting the same targets (1 to 28 day ahead incident hospitalizations for COVID-19).

The COVIDhub-baseline: A naive baseline model created and run by the US COVID-19 Forecast Hub meant to serve as a benchmark model to compare against. This model makes forecasts using the previous day’s incident hospitalization value as the point forecast (and 0.5 quantile for the probabilistic forecast) and past changes in daily incidence to calibrate uncertainty in its probabilistic forecasts. Beating the baseline is typically seen as one marker of a useful model [3]

The COVIDhub-4_week_ensemble: A median ensemble of all the submitting models for incident hospitalizations at the COVID-19 Forecast Hub, run by the Hub team that includes forecasts up to 28 day (4 week) ahead horizons. This model produces the same 1 to 14 day (1 to 2 week) ahead horizon forecasts as the COVIDhub-ensemble model, but also forecasts longer horizons. These longer horizons were removed from the main ensemble model due to research showing a sharp decline in accuracy in forecasts predicting beyond 2 week aheads (see Forecast horizons). However, these longer horizons are useful as a comparison for research purposes. Note that this ensemble model included slightly different contributing models based on what models were submitting to the COVID-19 Forecast Hub at a different times (check for specifics). [1], [6]

The forecasts for the comparison models are obtained from the Zoltar forecast archive, a research data repository developed by Reich Lab.

2.5 Temporal hierarchical forecasting (THieF)

- Add more mathematical detail here - Aggregation: We start with data at a daily scale. Then, based on the aggregation levels of interests for a particular model (e.g. weekly, every two weeks, every four weeks, etc), we induce this sort of artificial aggregation so that we have vectors of data add each of these time scales. Next, forecasts are made using each of these datasets separately. The reason that we are interested in different time scales is that there may be important patterns that are lost add higher time scales amid the noise of lower ones. More often than not, different patterns will be picked up at different aggregation levels, which means we get forecasts for the same future time periods that may not agree. This is where reconciliation comes into play, in which we combine all of the forecasts that we made.

2.6 Model specifications

2.6.1 THieF models

2.6.2 THieF ensemble models

2.6.3 ARIMA and SARIMA models

2.7 Metrics and Evaluation

2.7.1 Validation phase

2.7.2 Testing phase

CHAPTER 3

RESULTS

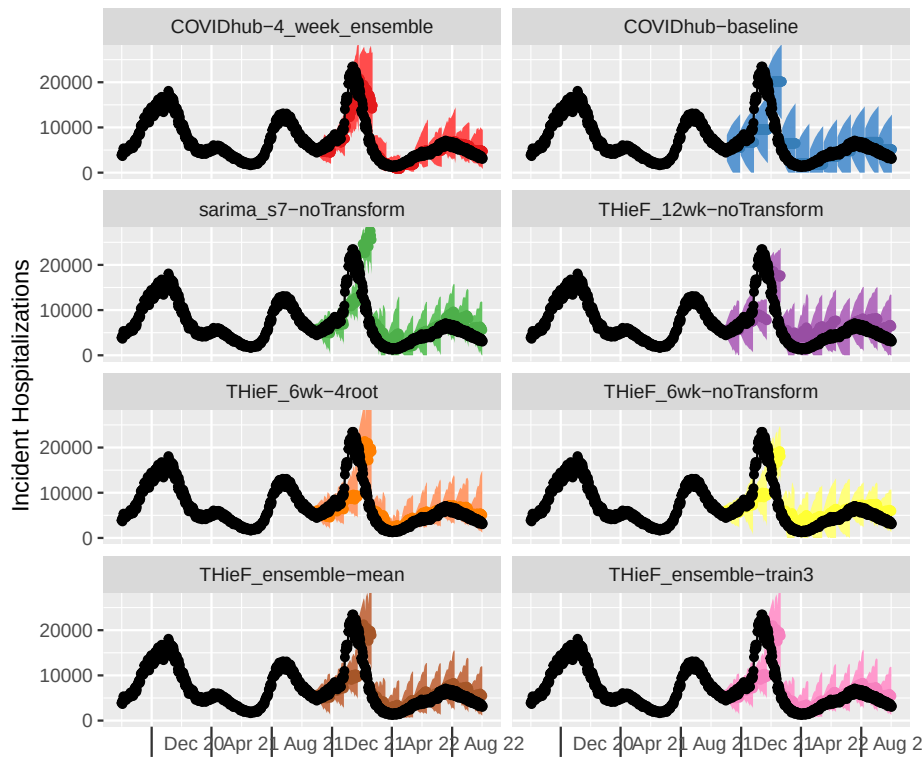
3.1 Validation phase

3.2 Testing phase

Daily COVID-19 Incident Hospitalizations: observed and forecasted

Selected location(s): United States

Selected forecast date(s): 2021-10-30, 2021-11-27, 2021-12-25, 2022-01-2



ensemble-train3, THieF_ensemble-mean, COVIDhub-baseline, COVIDhub-4_week_ensemble (forecasts)

Figure 3.1.

3.2.1 Summary of comparison models

3.2.2 Overall model performance

	model	wis	mae	cov_50	cov_95	rwis	rmae
1	COVIDhub-4_week_ensemble	1401.30	2085.10	0.63	0.91	0.75	0.83
2	THieF_6wk-4root	1642.50	2358.80	0.40	0.82	0.88	0.94
3	THieF_ensemble-train3	1687.40	2443.40	0.36	0.83	0.90	0.97
4	THieF_ensemble-mean	1700.50	2458.90	0.39	0.84	0.91	0.98
5	THieF_6wk-noTransform	1727.70	2504.80	0.46	0.86	0.92	1.00
6	COVIDhub-baseline	1867.50	2514.10	0.71	0.85	1.00	1.00
7	THieF_12wk-noTransform	2074.10	2908.30	0.37	0.86	1.11	1.16
8	sarima_s7-noTransform	2316.80	3042.40	0.41	0.80	1.24	1.21

Table 3.1. Summary of overall model performance for the US national level, aggregated across forecast week and 1 to 4 week ahead horizons. Prediction interval (PI) coverage rates measure the fraction of times the 50% and 95% PIs covered the true values. A well-calibrated model should achieve close to nominal (0.5 and 0.95, respectively) coverage rates. The “rwis” and “rmae” show relative weighted interval score and relative mean absolute error, which compare each model to the baseline. The baseline is defined to have relative metrics of 1. Models that perform better than the baseline have relative WIS or MAE less than 1. The table is ordered by ascending WIS value. Thus two THieF-4wk models show the best performance out of the THieF models, meeting the baseline for both relative WIS and MAE.

3.2.3 Model performance by horizon week

3.2.4 Model performance by pandemic wave

3.2.5 Model performance by location

3.2.6 Model performance by forecast week

	model	wis	mae	cov_50	cov_95	rwis	rmae
1	COVIDhub-4_week_ensemble	30.18	44.87	0.64	0.94	0.72	0.79
2	THieF_6wk-4root	34.55	50.12	0.51	0.90	0.83	0.88
3	THieF_ensemble-train3	35.62	51.57	0.56	0.91	0.86	0.91
4	THieF_ensemble-mean	35.91	51.80	0.57	0.91	0.86	0.91
5	THieF_6wk-noTransform	39.51	56.87	0.55	0.88	0.95	1.00
6	COVIDhub-baseline	41.63	56.77	0.79	0.96	1.00	1.00
7	THieF_12wk-noTransform	41.66	60.97	0.51	0.88	1.00	1.07
8	sarima_s7-noTransform	43.54	58.94	0.60	0.87	1.05	1.04

Table 3.2. Summary of overall model performance averaged over the states level. A similar ordering of models is seen, with the THieF-4wk-4root model performing the best and the THieF-12wk models performing the worst. However, the THieF-8wk models perform better in terms of model ranking for the states level. Additionally, relative WIS sees smaller values and 50% and 95% PI coverage see values closer to the nominal levels compared to that at the US national level, although relative MAE is larger with every model performing worse than the baseline.

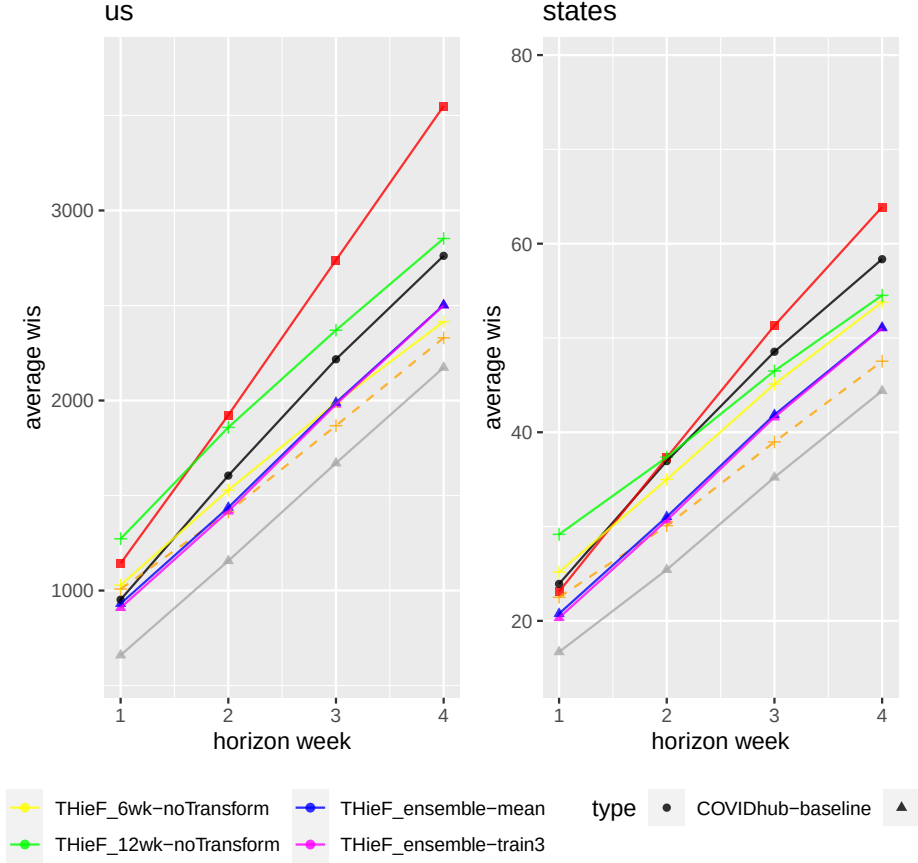


Figure 3.2. Average WIS by the horizon week for each model for the US national level and the states level. The figures show similar trends in WIS when forecasts are stratified by horizon week compared to being aggregated across horizon weeks. All models show worse performance relative to the baseline at the 1-week ahead horizon, likely a result of being overconfident at this horizon. Once again, the THieF-4wk-4root model performs the best for WIS. Plots for average MAE are not shown due to their similarity to the average WIS plots.

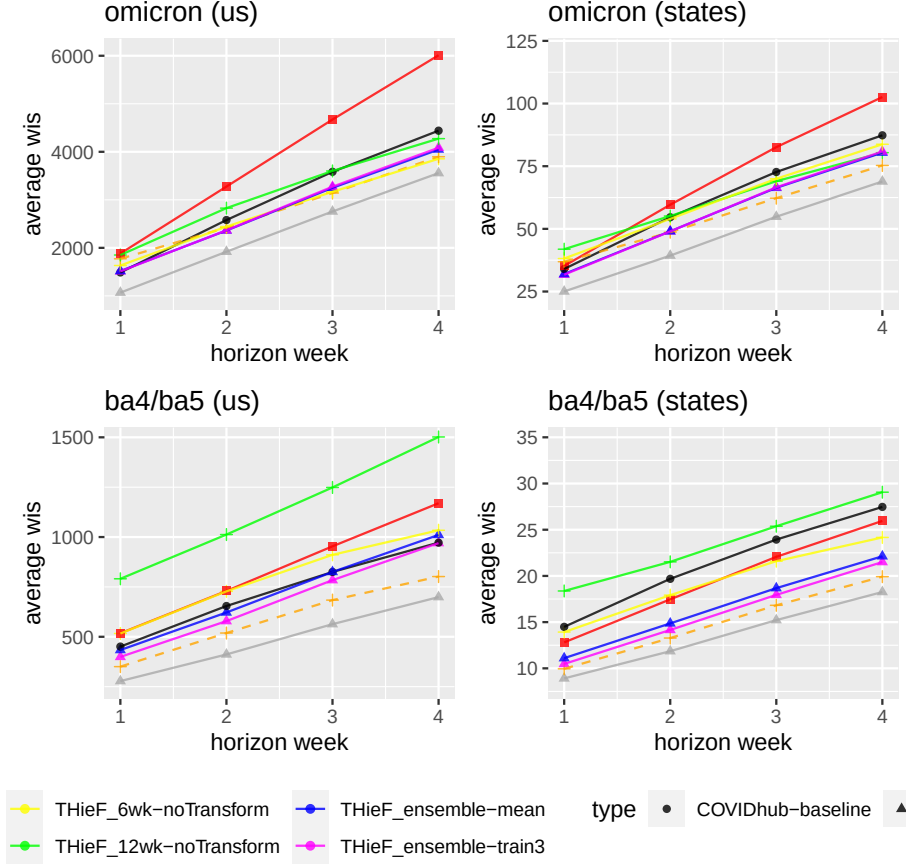


Figure 3.3. Average WIS by the horizon week and pandemic wave for each model for the US national level and the states level. The figures show similar trends in WIS when forecasts are stratified by horizon week and pandemic wave compared to being stratified only across horizon week. Again, all models show worse performance relative to the baseline at the 1-week ahead horizon likely due to being overconfident at this horizon. Model performance is more similar among the top four models during the Alpha wave compared to during the Winter21 wave. Notably, although the THieF-4wk-4root models largely show the lowest WIS, longer horizons during the Winter21 wave saw better performance by the poor-performing THieF-12wk models. Plots for average MAE are not shown due to their similarity to the average WIS plots.

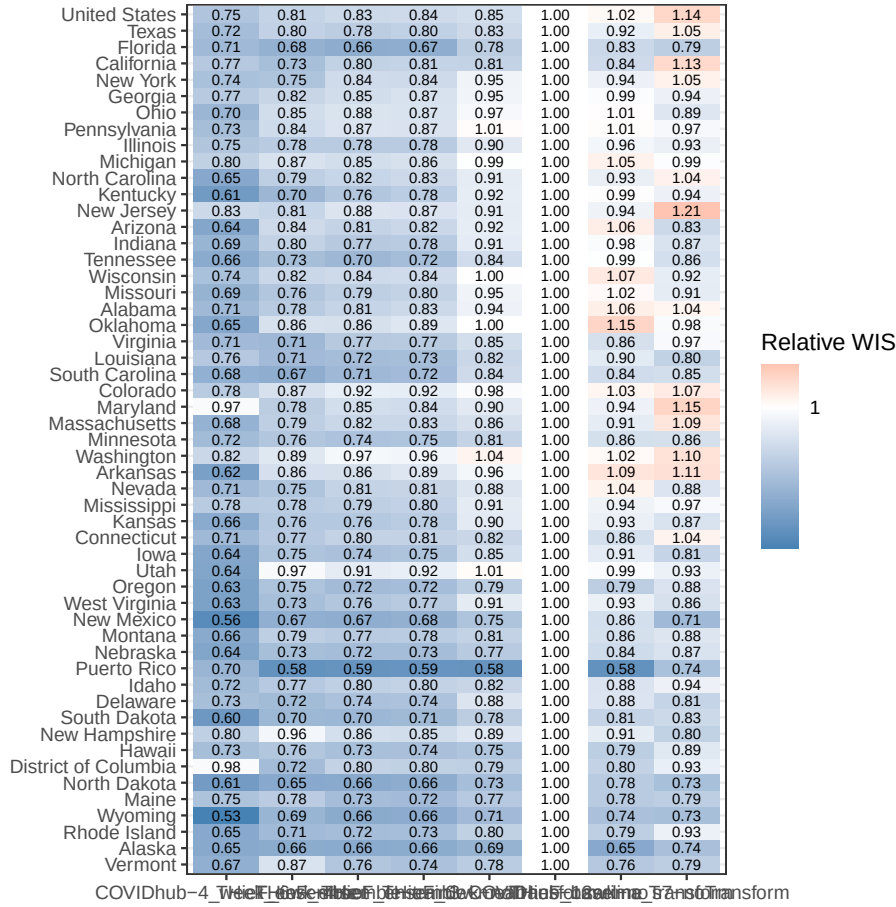


Figure 3.4. Relative WIS by location for each model across all horizons for the entire testing phase. Each square contains the average relative WIS of a model at a particular location, colored based on the relative WIS compare to the baseline. Blue indicates model outperformed the baseline for that location on average, while red indicates the model performed worse than the baseline. Locations are sorted based on cumulative hospitalizations during the testing phase while models are sorted from lowest to highest overall relative WIS, aggregated across states (Table 2). The top three models by relative WIS overall during the testing phase outperformed the baseline in every location.

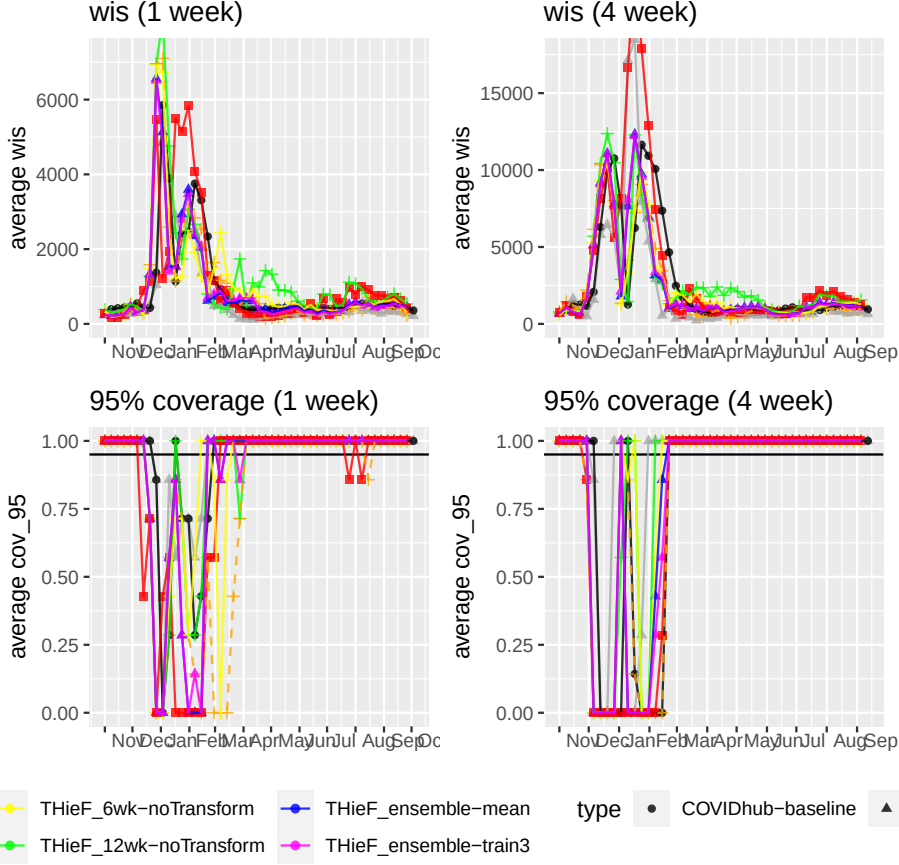


Figure 3.5. Average h-week ahead WIS and 95% PI coverage for each model for the US national level. Model performance differs from week to week: the WIS is the lowest during weeks with stable and low incident hospitalization values. The THieF-4wk models show the best performance with lower or comparable WIS to the baseline during the majority of the training period with especially good performance for 1-week ahead horizons. Meanwhile, the THieF-12wk showed lower or comparable WIS to the baseline during high incidence and changed points, which have been historically found difficult to forecast accurately. Plots for h-week ahead MAE are not shown due to their similarity to the WIS plots. All THieF models show very variable coverage rates, especially for a 1 week ahead horizon, only the models without a data transformation showing stable coverage starting around February 2021.

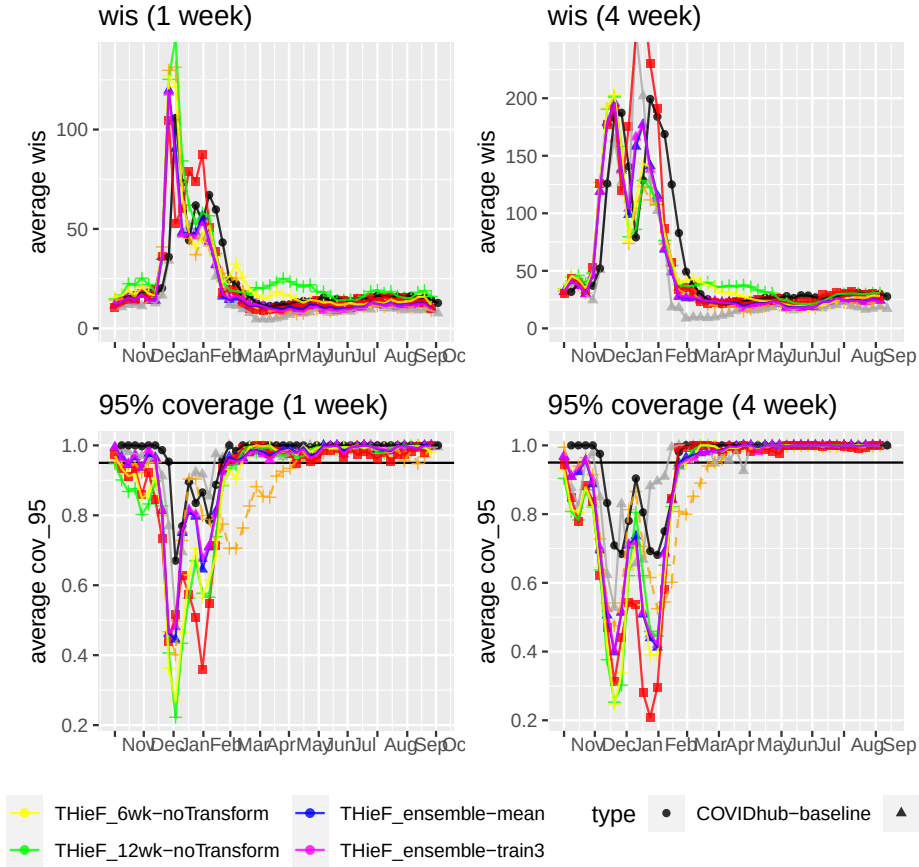


Figure 3.6. Average h-week ahead WIS and 95% PI coverage for each model for the states level. Once again, model performance differs from week to week with the lowest WIS values during weeks with stable and low incident hospitalization values and the THieF-4wk models showing the best performance. However, the THieF-12wk did not show good performance during high incidence and changed points for the states level; instead, THieF-12-4root actually showed some strangely large WIS values during April and May 2021. Plots for h-week ahead MAE are not shown due to their similarity to the WIS plots. One-week ahead coverage rates generally improved as time went on, the best models gaining stable and fairly good coverage rates starting in February 2021. For the 4-week ahead horizon, models without a data transformation showed the best performance and stable coverage starting around February 2021, though THieF-4wk-4root and THieF-8wk-4root also saw decent and fairly stable coverage rates after the downward Winter 21 change point.

CHAPTER 4

DISCUSSION

APPENDIX: SUPPLEMENTAL MATERIALS

	model	horizon_wk	wis	mae	cov_50	cov_95	rwis	rmae
1	COVIDhub-4_week_ensemble	1.00	659.80	966.80	0.68	0.94	0.69	0.77
2	THieF_ensemble-train3	1.00	909.60	1348.80	0.37	0.86	0.95	1.07
3	THieF_ensemble-mean	1.00	930.90	1372.90	0.38	0.86	0.98	1.09
4	COVIDhub-baseline	1.00	952.20	1258.80	0.73	0.92	1.00	1.00
5	THieF_6wk-4root	1.00	1007.40	1419.10	0.45	0.79	1.06	1.13
6	THieF_6wk-noTransform	1.00	1028.70	1467.40	0.46	0.88	1.08	1.17
7	sarima_s7-noTransform	1.00	1143.40	1533.90	0.32	0.81	1.20	1.22
8	THieF_12wk-noTransform	1.00	1272.40	1793.60	0.40	0.88	1.34	1.42
9	COVIDhub-4_week_ensemble	2.00	1156.20	1694.70	0.66	0.92	0.72	0.80
10	THieF_6wk-4root	2.00	1413.20	2029.00	0.42	0.85	0.88	0.90
11	THieF_ensemble-train3	2.00	1417.10	2046.90	0.39	0.84	0.88	0.97
12	THieF_ensemble-mean	2.00	1436.90	2070.80	0.40	0.85	0.90	0.98
13	THieF_6wk-noTransform	2.00	1528.50	2177.00	0.47	0.86	0.95	1.03
14	COVIDhub-baseline	2.00	1604.90	2113.30	0.74	0.85	1.00	1.00
15	THieF_12wk-noTransform	2.00	1858.50	2558.30	0.38	0.85	1.16	1.23
16	sarima_s7-noTransform	2.00	1923.10	2536.30	0.41	0.82	1.20	1.20
17	COVIDhub-4_week_ensemble	3.00	1670.60	2528.80	0.60	0.90	0.75	0.84
18	THieF_6wk-4root	3.00	1867.40	2715.30	0.37	0.84	0.84	0.93
19	THieF_ensemble-train3	3.00	1979.60	2845.10	0.35	0.81	0.89	0.93
20	THieF_6wk-noTransform	3.00	1989.00	2883.60	0.46	0.85	0.90	0.90
21	THieF_ensemble-mean	3.00	1989.00	2860.10	0.39	0.83	0.90	0.93
22	COVIDhub-baseline	3.00	2217.00	2993.70	0.71	0.83	1.00	1.00
23	THieF_12wk-noTransform	3.00	2369.90	3313.80	0.37	0.86	1.07	1.13
24	sarima_s7-noTransform	3.00	2738.40	3581.40	0.46	0.80	1.24	1.20
25	COVIDhub-4_week_ensemble	4.00	2173.40	3232.70	0.56	0.89	0.79	0.83
26	THieF_6wk-4root	4.00	2330.00	3341.70	0.36	0.80	0.84	0.88
27	THieF_6wk-noTransform	4.00	2414.90	3567.40	0.43	0.85	0.88	0.94
28	THieF_ensemble-train3	4.00	2501.10	3615.30	0.32	0.80	0.91	0.90
29	THieF_ensemble-mean	4.00	2502.20	3613.60	0.38	0.80	0.91	0.93
30	COVIDhub-baseline	4.00	2761.40	3782.10	0.67	0.80	1.00	1.00
31	THieF_12wk-noTransform	4.00	2852.50	4049.10	0.31	0.86	1.03	1.07
32	sarima_s7-noTransform	4.00	3549.40	4630.40	0.45	0.76	1.28	1.22

Table 1. Summary of model performance stratified by horizon week for the US national level, aggregated across forecast week. The results from this table were used to create Figure 4 that summarized average WIS by horizon week for the US national level and the states level. The table is ordered by ascending horizon week and WIS, with THieF-4wk-4root having the lowest WIS except for the 4 week ahead horizon. The baseline consistently does not have the best WIS.

	model	horizon_wk	wis	mae	cov_50	cov_95	rwis	rmae
1	COVIDhub-4_week_ensemble	1.00	16.70	25.11	0.65	0.96	0.70	0.79
2	THieF_ensemble-train3	1.00	20.32	30.04	0.55	0.93	0.85	0.94
3	THieF_ensemble-mean	1.00	20.76	30.52	0.56	0.93	0.87	0.96
4	THieF_6wk-4root	1.00	22.50	32.97	0.49	0.89	0.94	1.03
5	sarima_s7-noTransform	1.00	23.11	31.92	0.57	0.89	0.97	1.00
6	COVIDhub-baseline	1.00	23.92	31.87	0.80	0.97	1.00	1.00
7	THieF_6wk-noTransform	1.00	25.17	36.41	0.52	0.89	1.05	1.14
8	THieF_12wk-noTransform	1.00	29.17	42.20	0.49	0.89	1.22	1.32
9	COVIDhub-4_week_ensemble	2.00	25.42	37.90	0.66	0.95	0.69	0.77
10	THieF_6wk-4root	2.00	30.10	43.94	0.52	0.91	0.81	0.89
11	THieF_ensemble-train3	2.00	30.66	44.49	0.56	0.91	0.83	0.90
12	THieF_ensemble-mean	2.00	31.04	44.80	0.58	0.91	0.84	0.90
13	THieF_6wk-noTransform	2.00	34.99	50.27	0.56	0.88	0.95	1.02
14	COVIDhub-baseline	2.00	36.95	49.48	0.80	0.96	1.00	1.00
15	sarima_s7-noTransform	2.00	37.32	50.28	0.60	0.87	1.01	1.02
16	THieF_12wk-noTransform	2.00	37.35	54.31	0.53	0.88	1.01	1.10
17	COVIDhub-4_week_ensemble	3.00	35.22	52.52	0.64	0.94	0.73	0.79
18	THieF_6wk-4root	3.00	38.98	56.36	0.52	0.90	0.80	0.85
19	THieF_ensemble-train3	3.00	41.60	59.88	0.56	0.90	0.86	0.90
20	THieF_ensemble-mean	3.00	41.85	60.07	0.58	0.90	0.86	0.91
21	THieF_6wk-noTransform	3.00	45.11	64.83	0.56	0.87	0.93	0.98
22	THieF_12wk-noTransform	3.00	46.50	68.16	0.52	0.87	0.96	1.03
23	COVIDhub-baseline	3.00	48.54	66.30	0.79	0.95	1.00	1.00
24	sarima_s7-noTransform	3.00	51.31	69.07	0.60	0.86	1.06	1.04
25	COVIDhub-4_week_ensemble	4.00	44.39	65.43	0.63	0.93	0.76	0.80
26	THieF_6wk-4root	4.00	47.53	68.48	0.52	0.88	0.81	0.84
27	THieF_ensemble-train3	4.00	51.04	73.47	0.55	0.89	0.88	0.90
28	THieF_ensemble-mean	4.00	51.11	73.39	0.56	0.88	0.88	0.90
29	THieF_6wk-noTransform	4.00	53.80	77.46	0.55	0.86	0.92	0.95
30	THieF_12wk-noTransform	4.00	54.52	80.61	0.50	0.87	0.93	0.99
31	COVIDhub-baseline	4.00	58.35	81.23	0.77	0.95	1.00	1.00
32	sarima_s7-noTransform	4.00	63.89	86.44	0.60	0.85	1.09	1.06

Table 2. Summary of model performance stratified by horizon week for the states level, aggregated across forecast week. The results from this table were used to create Figure 4 that summarized average WIS by horizon week for the US national level and the states level. The table is ordered by ascending horizon week and WIS, with THieF-4wk-4root having the lowest WIS except for the 3 week (although it is second lowest) and 4 week ahead horizon. The baseline again consistently does not have the best WIS, performing progressively worse compared to the other models for each additional week ahead horizon.

	model	horizon_wk	wave	wis	mae	cov_50	cov_95	
1	COVIDhub-4_week_ensemble	1.00	ba4_ba5	277.80	378.70	0.89	1.00	
2	THieF_6wk-4root	1.00	ba4_ba5	350.40	542.90	0.60	0.99	
3	THieF_ensemble-train3	1.00	ba4_ba5	399.10	685.30	0.46	1.00	
4	THieF_ensemble-mean	1.00	ba4_ba5	433.60	740.30	0.49	1.00	
5	COVIDhub-baseline	1.00	ba4_ba5	450.80	486.90	0.92	1.00	
6	THieF_6wk-noTransform	1.00	ba4_ba5	516.50	822.50	0.60	1.00	
7	sarima_s7-noTransform	1.00	ba4_ba5	517.80	828.60	0.41	0.99	
8	THieF_12wk-noTransform	1.00	ba4_ba5	790.70	1319.20	0.35	1.00	
9	COVIDhub-4_week_ensemble	2.00	ba4_ba5	411.40	572.50	0.85	1.00	
10	THieF_6wk-4root	2.00	ba4_ba5	518.80	813.00	0.61	1.00	
11	THieF_ensemble-train3	2.00	ba4_ba5	578.60	966.60	0.49	1.00	
12	THieF_ensemble-mean	2.00	ba4_ba5	621.80	1030.30	0.50	1.00	
13	COVIDhub-baseline	2.00	ba4_ba5	654.20	741.00	0.98	1.00	
14	THieF_6wk-noTransform	2.00	ba4_ba5	727.80	1145.90	0.59	1.00	
15	sarima_s7-noTransform	2.00	ba4_ba5	729.70	1169.60	0.53	1.00	
16	THieF_12wk-noTransform	2.00	ba4_ba5	1012.60	1677.50	0.38	1.00	
17	COVIDhub-4_week_ensemble	3.00	ba4_ba5	563.30	824.70	0.77	1.00	
18	THieF_6wk-4root	3.00	ba4_ba5	683.90	1069.10	0.59	1.00	
19	THieF_ensemble-train3	3.00	ba4_ba5	784.00	1311.50	0.45	1.00	
20	COVIDhub-baseline	3.00	ba4_ba5	825.00	1038.90	1.00	1.00	
21	THieF_ensemble-mean	3.00	ba4_ba5	825.00	1371.30	0.50	1.00	
22	THieF_6wk-noTransform	3.00	ba4_ba5	911.00	1464.40	0.59	1.00	
23	sarima_s7-noTransform	3.00	ba4_ba5	953.50	1514.20	0.57	1.00	
24	THieF_12wk-noTransform	3.00	ba4_ba5	1248.70	2097.10	0.38	1.00	
25	COVIDhub-4_week_ensemble	4.00	ba4_ba5	698.90	1034.30	0.77	1.00	
26	THieF_6wk-4root	4.00	ba4_ba5	801.90	1210.30	0.64	1.00	
27	THieF_ensemble-train3	4.00	ba4_ba5	969.80	1585.10	0.48	1.00	
28	COVIDhub-baseline	4.00	ba4_ba5	972.00	1320.50	1.00	1.00	
29	THieF_ensemble-mean	4.00	ba4_ba5	1010.80	1651.70	0.53	1.00	
30	THieF_6wk-noTransform	4.00	ba4_ba5	1034.00	1631.70	0.61	1.00	
31	sarima_s7-noTransform	4.00	ba4_ba5	1169.60	1821.80	0.60	1.00	
32	THieF_12wk-noTransform	4.00	ba4_ba5	1501.80	2581.80	0.34	1.00	
33	THieF_ensemble-mean	1.00	delta	239.10	305.40	0.86	1.00	
34	THieF_ensemble-train3	1.00	delta	247.90	326.80	0.86	1.00	
35	sarima_s7-noTransform	1.00	delta	270.90	454.30	0.29	1.00	
36	THieF_6wk-4root	1.00	delta	327.60	536.40	0.71	1.00	
37	THieF_6wk-noTransform	1.00	delta	332.50	563.70	0.43	1.00	
38	THieF_12wk-noTransform	1.00	delta	381.80	523.00	1.00	1.00	
39	THieF_6wk-noTransform	2.00	delta	286.50	260.10	1.00	1.00	
40	sarima_s7-noTransform	2.00	delta	290.70	240.00	1.00	1.00	
41	THieF_ensemble-mean	2.00	delta	313.60	215.30	1.00	1.00	
42	THieF_ensemble-train3	2.00	delta	316.20	207.60	1.00	1.00	
43	THieF_6wk-4root	2.00	delta	324.40	280.40	1.00	1.00	
44	THieF_12wk-noTransform	2.00	delta	487.90	404.70	1.00	1.00	
45	THieF_6wk-noTransform	3.00	delta	396.00	511.10	0.86	1.00	
46	sarima_s7-noTransform	19 3.00	delta	486.30	701.80	1.00	1.00	
47	THieF_ensemble-mean	3.00	delta	509.60	800.20	1.00	1.00	
48	THieF_ensemble-train3	3.00	delta	509.60	797.20	1.00	1.00	

	model	horizon_wk	wave	wis	mae	cov_50	cov_95	rw
1	COVIDhub-4_week_ensemble	1.00	ba4_ba5	8.91	13.43	0.72	0.99	0.6
2	THieF_6wk-4root	1.00	ba4_ba5	9.95	14.83	0.62	0.97	0.6
3	THieF_ensemble-train3	1.00	ba4_ba5	10.48	15.62	0.67	0.99	0.7
4	THieF_ensemble-mean	1.00	ba4_ba5	11.10	16.47	0.69	0.99	0.7
5	sarima_s7-noTransform	1.00	ba4_ba5	12.81	19.00	0.68	0.98	0.8
6	THieF_6wk-noTransform	1.00	ba4_ba5	13.93	20.54	0.70	0.99	0.9
7	COVIDhub-baseline	1.00	ba4_ba5	14.49	16.59	0.92	1.00	1.0
8	THieF_12wk-noTransform	1.00	ba4_ba5	18.37	28.32	0.61	0.99	1.2
9	COVIDhub-4_week_ensemble	2.00	ba4_ba5	11.85	17.64	0.76	1.00	0.6
10	THieF_6wk-4root	2.00	ba4_ba5	13.28	19.07	0.69	0.99	0.6
11	THieF_ensemble-train3	2.00	ba4_ba5	14.15	20.34	0.74	0.99	0.7
12	THieF_ensemble-mean	2.00	ba4_ba5	14.86	21.20	0.75	0.99	0.7
13	sarima_s7-noTransform	2.00	ba4_ba5	17.46	25.07	0.75	0.99	0.8
14	THieF_6wk-noTransform	2.00	ba4_ba5	17.92	25.69	0.77	1.00	0.9
15	COVIDhub-baseline	2.00	ba4_ba5	19.69	21.67	0.97	1.00	1.0
16	THieF_12wk-noTransform	2.00	ba4_ba5	21.52	32.82	0.68	1.00	1.0
17	COVIDhub-4_week_ensemble	3.00	ba4_ba5	15.22	22.86	0.76	0.99	0.6
18	THieF_6wk-4root	3.00	ba4_ba5	16.82	23.30	0.72	0.99	0.7
19	THieF_ensemble-train3	3.00	ba4_ba5	17.95	25.29	0.76	1.00	0.7
20	THieF_ensemble-mean	3.00	ba4_ba5	18.68	26.22	0.78	1.00	0.7
21	THieF_6wk-noTransform	3.00	ba4_ba5	21.59	31.20	0.78	1.00	0.9
22	sarima_s7-noTransform	3.00	ba4_ba5	22.06	31.34	0.78	0.99	0.9
23	COVIDhub-baseline	3.00	ba4_ba5	23.93	27.49	0.97	1.00	1.0
24	THieF_12wk-noTransform	3.00	ba4_ba5	25.38	39.28	0.70	1.00	1.0
25	COVIDhub-4_week_ensemble	4.00	ba4_ba5	18.26	27.51	0.76	0.99	0.6
26	THieF_6wk-4root	4.00	ba4_ba5	19.92	26.57	0.75	1.00	0.7
27	THieF_ensemble-train3	4.00	ba4_ba5	21.50	29.63	0.78	1.00	0.7
28	THieF_ensemble-mean	4.00	ba4_ba5	22.13	30.41	0.79	1.00	0.8
29	THieF_6wk-noTransform	4.00	ba4_ba5	24.17	34.33	0.80	1.00	0.8
30	sarima_s7-noTransform	4.00	ba4_ba5	25.95	36.72	0.80	0.99	0.9
31	COVIDhub-baseline	4.00	ba4_ba5	27.46	32.80	0.97	1.00	1.0
32	THieF_12wk-noTransform	4.00	ba4_ba5	29.06	45.41	0.69	1.00	1.0
33	sarima_s7-noTransform	1.00	delta	10.38	14.69	0.70	0.97	0.3
34	THieF_ensemble-mean	1.00	delta	11.60	18.76	0.64	0.99	0.3
35	THieF_ensemble-train3	1.00	delta	11.87	19.30	0.60	0.99	0.3
36	THieF_6wk-4root	1.00	delta	14.16	22.68	0.51	0.95	0.4
37	THieF_12wk-noTransform	1.00	delta	14.89	23.52	0.52	0.95	0.4
38	THieF_6wk-noTransform	1.00	delta	15.57	25.83	0.51	0.97	0.4
39	sarima_s7-noTransform	2.00	delta	14.64	20.80	0.69	0.98	0.2
40	THieF_ensemble-mean	2.00	delta	16.33	26.27	0.65	0.99	0.3
41	THieF_ensemble-train3	2.00	delta	16.52	26.76	0.64	0.98	0.3
42	THieF_6wk-4root	2.00	delta	17.99	29.88	0.54	0.98	0.3
43	THieF_12wk-noTransform	2.00	delta	19.35	30.10	0.60	0.95	0.3
44	THieF_6wk-noTransform	2.00	delta	19.58	32.20	0.62	0.97	0.3
45	sarima_s7-noTransform	3.00	delta	23.20	35.71	0.60	0.97	0.3
46	THieF_ensemble-mean	20 3.00	delta	25.61	41.49	0.58	0.97	0.3
47	THieF_ensemble-train3	3.00	delta	25.77	41.94	0.57	0.98	0.3
48	THieF_6wk-4root	3.00	delta	27.03	44.49	0.52	0.99	0.3

BIBLIOGRAPHY

- [1] COVID-19 Forecast Hub. Ensemble model. <https://covid19forecasthub.org/doc/ensemble/>. Accessed: 2023-03-15.
- [2] Cramer, Estee Y, Huang, Yuxin, Wang, Yijin, Ray, Evan L, Cornell, Matthew, Bracher, Johannes, Brennen, Andrea, Castro Rivadeneira, Alvaro J, Gerding, Aaron, House, Katie, Jayawardena, Dasuni, Kanji, Abdul H, Khandelwal, Ayush, Le, Khoa, Niemi, Jarad, Stark, Ariane, Shah, Apurv, Wattanachit, Nutch, Zorn, Martha W, Reich, Nicholas G, and Consortium, US COVID-19 Forecast Hub. The united states covid-19 forecast hub dataset. *medRxiv* (2021).
- [3] Cramer, Estee Y., Ray, Evan L., Lopez, Velma K., Bracher, Johannes, Brennen, Andrea, Rivadeneira, Alvaro J. Castro, Gerding, Aaron, Gneiting, Tilmann, Katie H. House, Yuxin Huang, Dasuni Jayawardena, Kanji, Abdul H., Khandelwal, Ayush, Le, Khoa, Mühlemann, Anja, Niemi, Jarad, Shah, Apurv, Stark, Ariane, Wang, Yijin, Wattanachit, Nutch, ..., and Reich, Nicholas G. Evaluation of individual and ensemble probabilistic forecasts of covid-19 mortality in the us. *Proceedings of the National Academy of Sciences* 119, 15 (April 2022), e2113561119.
- [4] Donovan, Doug. Johns hopkins covid-19 data hub ends after three years. *The Hub* (March 2023).
- [5] Gibson, Graham C., Reich, Nicholas G., and Sheldon, Daniel. Real-time mechanistic bayesian forecasts of covid-19 mortality. *medRxiv : the preprint server for health sciences* (December 2020).
- [6] Reich, Nicholas G., Tibshirani, Ryan J., Ray, Evan L., and Rosenfeld, Roni. On the predictability of covid-19. <https://forecasters.org/blog/2021/09/28/on-the-predictability-of-covid-19/>, September 2021. Accessed: 2022-04-06.
- [7] U.S. Department of Health and Human Services. Covid-19 reported patient impact and hospital capacity by state timeseries. <https://healthdata.gov/Hospital/COVID-19-Reported-Patient-Impact-and-Hospital-Capa/g62h-syeh>. Accessed: 2023-03-15.